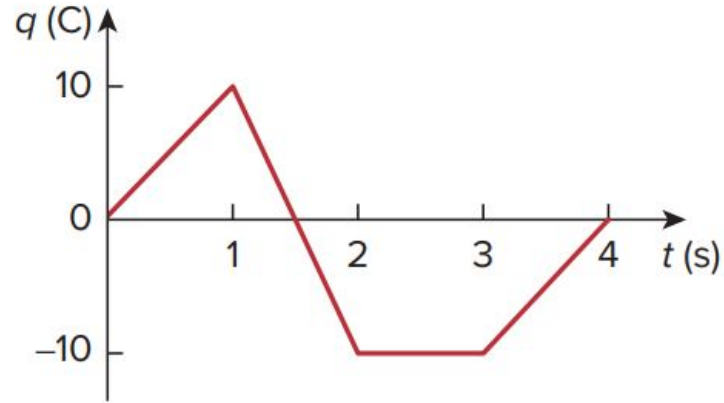


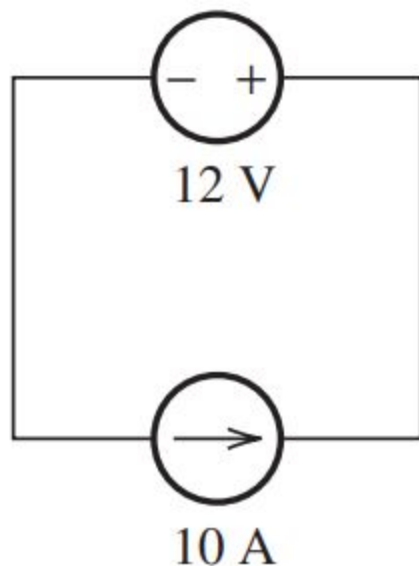
1. The charge flowing in a wire is plotted in Fig. 1 Sketch the corresponding current

Fig. 1.

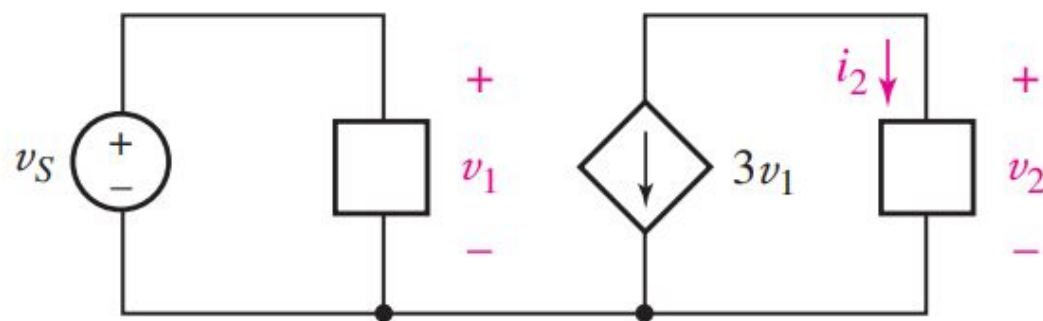


Hints: Obtain straight line equations of charge as function of time for different time interval and the equation for current.

P1.67. Given the circuit shown in Figure P1.67, find the power for each source. Which source is absorbing power? Which is delivering power?



35. The circuit depicted in Fig. 2.34 contains a dependent current source; the magnitude and direction of the current it supplies are directly determined by the voltage labeled v_1 . Note that therefore $i_2 = -3v_1$. Determine the voltage v_1 if $v_2 = 33i_2$ and $i_2 = 100$ mA.



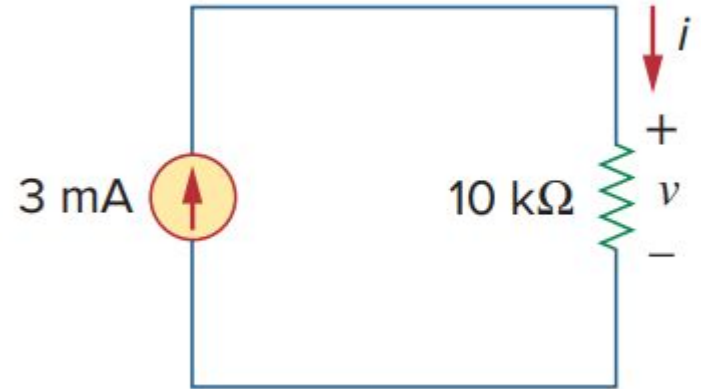
■ **FIGURE 2.34**

46. The following experimental data is acquired for an unmarked resistor, using a variable-voltage power supply and a current meter. The current meter readout is somewhat unstable, unfortunately, which introduces error into the measurement.

Voltage (V)	Current (mA)
−2.0	−0.89
−1.2	−0.47
0.0	0.01
1.0	0.44
1.5	0.70

- (a) Plot the measured current-versus-voltage characteristic.
- (b) Using a best-fit line, estimate the value of the resistance.

Determine the voltage v using Ohm's law.



How much resistance is required to limit the current to 1.5 mA if the potential drop across the resistor is 24 V?

If the current through a 200 k Ω resistor is 5.0 μ A, what is the voltage drop across the resistor?

- a.** If an electric heater draws 9.5 A when connected to a 120 V supply, what is the internal resistance of the heater?